

Methods In Global Distributed Tracking For Class I Medical Device Manufacturing

OVERVIEW

Global Distributed Tracking (GDT) is software developed by the non-profit Global Open Source Quality Assurance System (GOSQAS) that enables the tracking of physical objects via scannable QR codes linked to an encrypted, append-only provenance record [6][7]. Decentralized Medical Device Manufacturing (DMDM) is a community-scale, non-profit, FDA-registered Class I medical device manufacturer. DMDM has implemented GDT across manufacturing and post-market surveillance operations to fulfill several FDA requirements without the overhead of traditional quality management software or physical records. This whitepaper describes those implementations in detail, with the intent of providing a replicable methodology for other community-scale operations for whom quality system implementation represents a barrier.

Purpose

GDT has utility in small-scale manufacturing across both regulated and unregulated organizations, especially for decentralized aid operations. We intend to share the methods and justifications for our implementation of GDT to demonstrate possible use cases of GDT and to advance the conversation around quality management in the open source manufacturing community.

Scope

We will cover a general overview of DMDM's implementation of GDT for tool maintenance schedules, lot acceptance records, labeling, device identification and traceability, complaints, corrective and preventive actions (CAPA), reworks and recalls, as well as best practices for logistic tracking. DMDM is an FDA-registered manufacturing facility producing class 1 medical

devices. The scope of our implementation is limited to QMSR [2], which is largely based on ISO13485:2016 [4].

GLOBAL DISTRIBUTED TRACKING

GDT is an open source logistics tracking software tool that allows for authentication free tracking. It is similar to an anonymized postal tracking app in that it uses many different types of information packaging to allow for the tracking of one or multiple objects through time and space. This allows manufacturers to trace deliveries through at-risk communities, such as conflict zones, without disclosing information that may compromise the distributor or recipient [8]. GDT consists of the following features.

Records & Entries

Records are the base of GDT. Each object you want to track should probably have its own record, with a unique alphanumeric Record Key (RK). Anyone, and only someone, with that RK can access the record.

Once accessed you can add entries which allow for information to be added to the record. The entries record the exact time they were made and any new information the accesser wants. All entries are immutable and in series.

Groups

Group records can be generated where a parent RK is associated with an arbitrary number of child RKs. At present, only a single parent/child depth is possible, and multi-level functionality, such as a lot of crates of boxes of parts, will be provided by the Application Programmers Interface (API) as a future feature. These groups have two unique keys: 1) the “parent key” which allows for mass write actions across all child records and 2) the “reporting key” which acts as a normal record, used for group tracking and does not display or write to child records.

Tags

Tags are part of record entries and are intended to sort entries based on what information they are adding to the system, such as *‘deployed’*, *‘defective’*, *‘received’*. However, tags are free text and have applications beyond the intended use.

QR Codes

The various keys are accessible from the GOSQAS website through the URL generated. This URL is automatically translated into a QR code (Fig. 1) within GDT but can easily be generated through any QR code program. For simple integration, we use QR codes generated by python scripts for batch printing of customer facing labels.



Track this package!

Scan this QR code to anonymously get any important information added by people up the distribution chain, such as product recalls or shipping info.

Please add whatever you think might be necessary! Only the time and what you choose to say will be logged in the program.

Fig. 1. An example of a back package label with client facing instructions on using GDT.

QUALITY ASSURANCE

Tool Maintenance

Maintaining tool maintenance and calibration records is both a best practice and a regulatory requirement for DMDM [3]. Tool maintenance records are important for root cause analysis in the event of nonconforming products and demonstrating proper upkeep. Categorizing which tools and equipment need regular inspection and maintenance, and keeping up-to-date, easily accessed documentation of activities is an important quality assurance step for such devices.

For operations like DMDM, it is important that quality-related information, such as tool maintenance be accessible remotely. Global Distributed Tracking enables collaboration, communication, and scalability for global initiatives by providing quality assurance tracking of tool and equipment maintenance in a manner which reduces human error, enables easy auditing, and produces reliable record-keeping.

GDT meets regulatory standards by providing a system that leaves an immutable audit trail of activities and records. GDT allows for convenient record keeping by having a key in the form of a QR code attached to the tool, the operator can scan the tool at point of use to determine it's status and record any maintenance. A current regulatory limitation of GDT, given it's pseudo-anonymous data entry, is lack of compliance with FDA electronic signature requirements (21 CFR Part 11). This is a recurring limitation specific to FDA regulated facilities and outside the scope of GDT).

Fig. 2 provides a walkthrough of how to create a new record for tools and equipment in GDT:

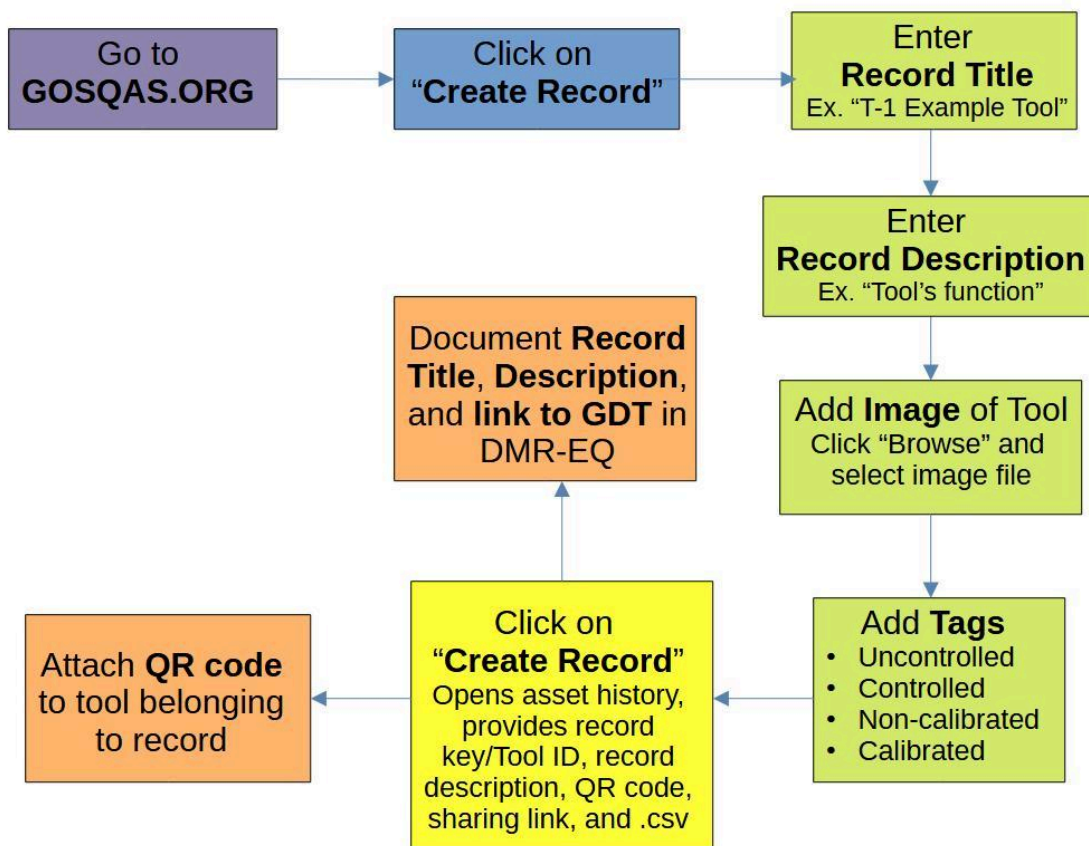


Fig. 2 - User walkthrough of how to create a GDT record for tool and equipment maintenance. Once a record creation is initiated (purple and blue), the tool details must be entered into the record information (green) before the new record is completed (yellow). Once a record is created, copies of the record information must be entered into the DMR-EQ as well as physically attached to the tool (orange).

Internal Record Keeping

DMDM maintains tabular data of every lot produced and its individual RKs. These keys are considered confidential and kept private to avoid potential disclosure of personal health information (PHI). Additionally, keeping the keys private prevents people from “graffitiing” records. These keys will, however, be printed on the individual distributed tourniquets. If a bad actor discovers an individual key, they can only append graffiti to that individual record.

The reporting key can be printed on the box of an entire lot, and can safely be used by a distributor to record the history of the entire lot until the lot is broken into smaller units for sale or redistribution, at which point the “reporting key” should be labeled “group distributed” and then abandoned in the field.

Acceptance Records

DMDM makes tourniquets in batches that we refer to as “lots”. DMDM conducts lot acceptance testing prior to the release of any product. To make these records maximally accessible to both the FDA and users we put them directly into GDT.

To demonstrate our standard of quality, we use the Western University Open Source Tourniquet Testing System (TRNARM), an open source device used for measuring the tourniquet’s application pressure enabling us to compare lot to lot and lot to predicate device performance [9]. We use AQL tables [5] for our sample plan, select tourniquets pseudo-randomly, and perform destructive testing on each. The results of this testing are recorded both in the individual keys of each tested tourniquet and publicly on our Codeburg repository.

For those adapting these methods, we recommend researching and adapting to current testing methods in your industry. AQL tables are proprietary, but the principles behind them are so widespread that the information can often be readily found.

Our tourniquets are used in life-threatening emergency situations, therefore our sampling must be aggressive. If any defects are found, the whole lot must be inspected and investigated thoroughly for correction. In the event of remanufacturing, the specific operations performed are recorded in each unit key.

LOGISTICS

Labels

All medical devices produced by DMDM have FDA-compliant labels which require some form of unique identification. This allows for accurate tracking of devices during recalls. Currently, Class I

devices require either a Unique Device Identifier (UDI) or a combination of a Universal Product Code (UPC) and a serial number [1].

GDT keys are excellent for this purpose, because not only do they provide a unique Identifiable number (fulfilling the code requirement), but also allow for the users to trace provenance of the device in their hands. By putting the QR code in multiple places, we can increase usage, allowing for better odds of usage.

All medical devices produced by DMDM have attached to them FDA-compliant labels with a Unique Device Identifier (UDI). For Class I devices, the requirement is a Device Identifier (DI), which identifies the product, registered in FDA's GUDID database. Serialization, meaning a unique identifier for each individual unit, is not required at the Class I level.

DMDM exceeds this requirement by shipping each device with a UDI on the packaging and a unique GDT key printed directly on the device itself, giving every unit its own scannable provenance record, note that the packaging UDI label is uploaded as part of each product key during group record creation in order to preserve the link in the event the packaging is discarded. GDT allows us to easily gain unit level records resolution. Where a standard UDI tells you what the device is, the GDT key tells the full history of the device.

Currently DMDM UDIs are obtained through FDA approved UDI issuers. If GOSQAS were to obtain a UDI issuing authority license, the GDT key itself could serve as the UDI, streamlining the labeling requirement into a single identifier.

Below are example labels currently used in production. Fig 3 is the package label, which contains the UPC and markings required by FDA. Fig 4 shows the label affixed to each unit, which contains the GDT record key QR code.

NON-PNEUMATIC TOURNIQUET

REF TQ1

LOT 3_250226



Store out of
direct
sunlight



Store in a cool dry
location—do not
exceed 60°C / 140°F



DMDM



Distributed
Medical Device
Manufacturing

657 W Saint Marys Rd,
Tucson, AZ 85701, USA
info@dmdm.icu

Fig. 3 - Example package label, note the use of a proprietary barcode for the UPC.

NON-PNEUMATIC TOURNIQUET TQ1

LOT 3_250226

Distributed Medical
Device Manufacturing
657 W St. Marys Rd,
Tucson, AZ 87501
<https://dmdm.icu>
info@dmdm.icu



Fig. 4 - Example unit label.

Identification and Traceability

The FDA requires manufacturers to keep Device History Records (DHR) for every lot of medical devices manufactured. This information is retained on file in case a problem occurs in a given lot, allowing people to review the acceptance information to learn about what issues are present, locate the tourniquets, and to develop corrective and preventive actions such as field actions and recalls.

The group function allows functionality stacking of all DHR information tagged to each individual device. This reduces the need to cross-reference device serial numbers between what is stored in group records and what is not.

What needs to go in a DHR is quite precise, so we illustrated the SOP for it below (Fig. 5):

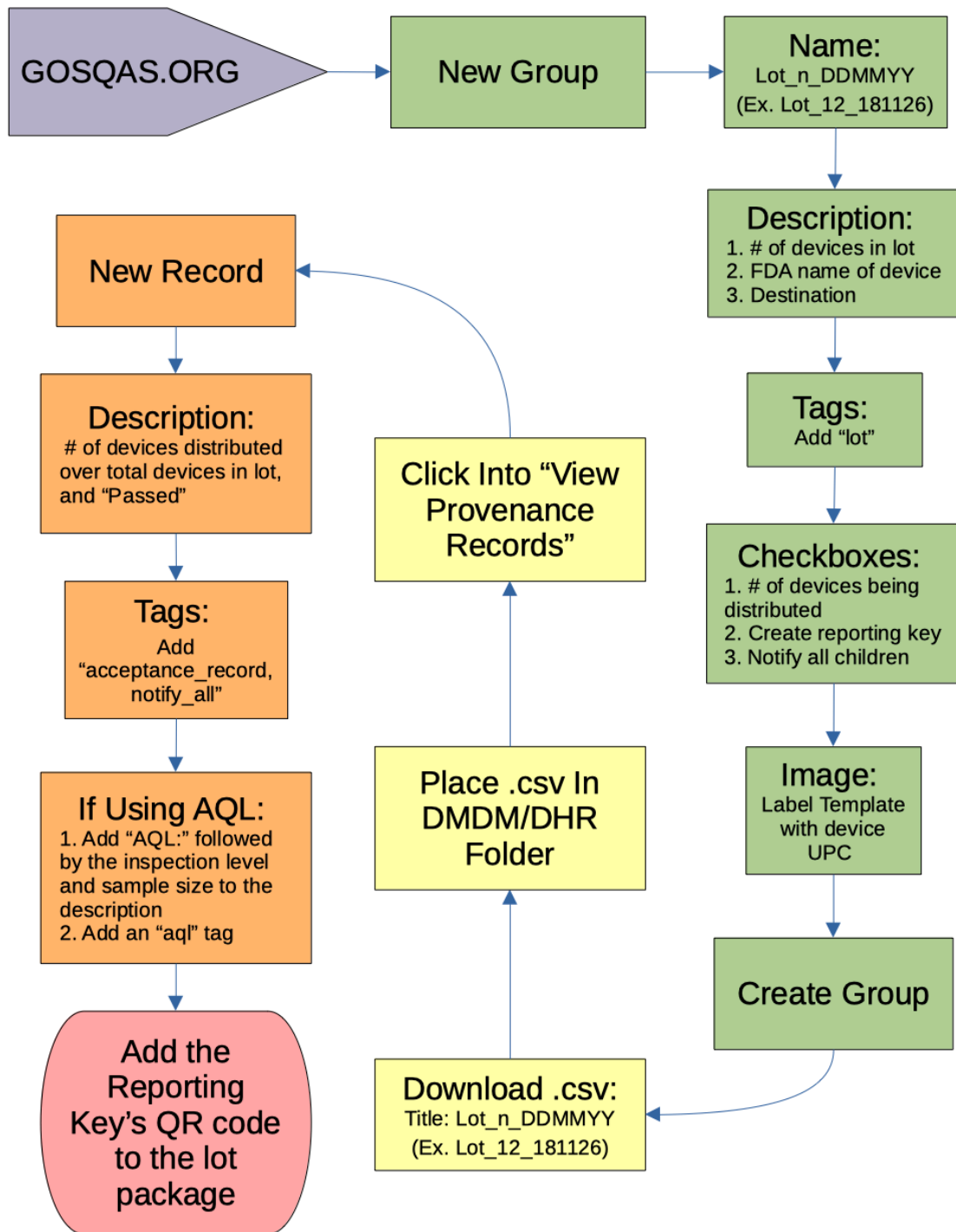


Fig. 5. After a lot has passed QA, this procedure must be followed to assign keys to each device and add the DHR information to GDT.

By encouraging our distributors to use GDT to trace our packages anonymously, we can trace provenance information even through conflict zones, without compromising group operational security. This way we can make sure our goods are getting where they need to and we are able to receive customer feedback without the need for direct communication.

REPORTING

Complaints

Complaints are a core part of any system that seeks to continuously improve and an integral part of FDA post-market surveillance requirements [1]. The feedback created by the devices being sent out in the world are an invaluable resource for tuning production to the best fit for users. For this reason, the FDA and ISO:13485 requires all complaints, formal and informal, to be documented and evaluated for reporting. Every identified complaint must either be investigated or have a strong justification for not investigating and monitoring for trends. A good faith effort must be made to obtain customer information, and details from the complainant.

This seems on its face a good use for GDT, as GDT allows communication from user to maker. However the problem is, how does one track every device from every lot and monitor each for potential complaints? There will be too much random noise from record history, successful usage information, and almost certainly some digital graffiti.

This use case is at the edge of what GDT can currently accomplish, however with GDT API access DMDM has developed a polling system to retrieve new information entered into all record keys for tracking and trending complaints in an external application. The function of this application is outside the scope of this paper and we expect to release further information or roll our functionality into the GDT project in the near future.

For a high level functional diagram see Fig. 6.

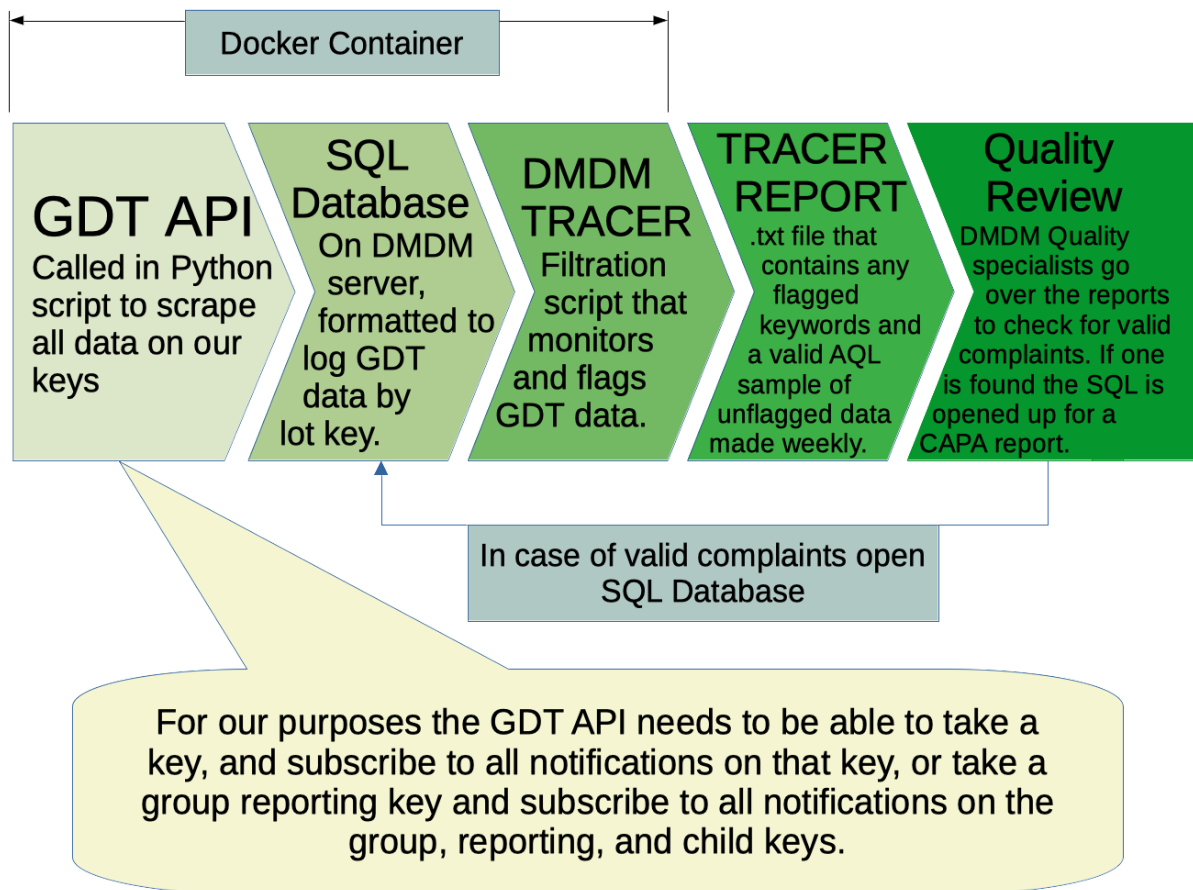


Fig. 6 - Proposed architecture of the FDA compliant complaint management system using GDT.

Corrective And Preventative Action and Recalls

Investigations into complaints are done via Corrective And Preventative Action (CAPA, which as stated above will be informed by information taken from GDT, but will be retained on file externally.

We tried to determine if GDT could be used in lieu of a CAPA record, but It was decided that there was too much information (text, photos and data), and too complex to live on the GDT record chain. Instead we cross link CAPA records maintained in our Codeburg record repository with individual unit record keys related to each CAPA. This allows us to easily track and trend individual units with issues and aggregate information required to complete the CAPA.

Because GDT has a built-in recall system, it's very simple to bring it up to FDA specifications. We simply add a new record entry to the group record, tag it as "recall", and include the reason for the recall in the description, the date we were made aware of the problem, the current shop

address for returns, and instructions for customer communication. This is potentially the most powerful use of GDT and why we exceed FDA requirements for individual UDI. Conducting a comprehensive recall in dispersed and chaotic aid operations across continents would essentially be impossible without a tool like GDT.

Conclusion

DMDM's implementation of GDT demonstrates that a single open source platform can satisfy multiple FDA documentation requirements: acceptance records, device history records, complaint management, recall notification, and tool maintenance logs. This reduces the complexity and cost of separate systems, dedicated software licenses, and administrative overhead. Implementing quality management systems in small scale manufacturing is challenging, GDT is free, requires no authentication, runs in a browser, and leaves an immutable, auditable provenance trail on every record it touches.

DMDM strives to not just meet, but exceed, FDA requirements in order to ensure our products are safe and effective for those who need them. Every device DMDM ships carries both a UDI and a unique GDT key, giving field operators, distributors, and end users direct access to the provenance history of the specific unit in their hands. The QR code on the device is itself the complaint intake mechanism: a user encountering a defective device can initiate a complaint record without navigating a separate reporting system.

The methods described in this paper are replicable. Any small scale manufacturer willing to invest the setup time can integrate GDT into their quality management system at zero software cost. Open source manufacturing deserves open source quality assurance, and GDT makes that available today. The GDT team is continually adding new features and refining the tool. Through our work implementing GDT we found the team to be incredibly responsive, responding to GitHub issues we submitted and answering questions. The more people that use GDT and provide feedback enables quicker and more directed development of the platform. To those reading this thinking about manufacturing 1 or 10000 devices, we would urge you to evaluate how GDT can be integrated into your process.

REFERENCES

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